

A Mechanistic Model for Radiation-Induced Segregation Controlled Grain-Boundary Oxidation in Austenitic Stainless Steels

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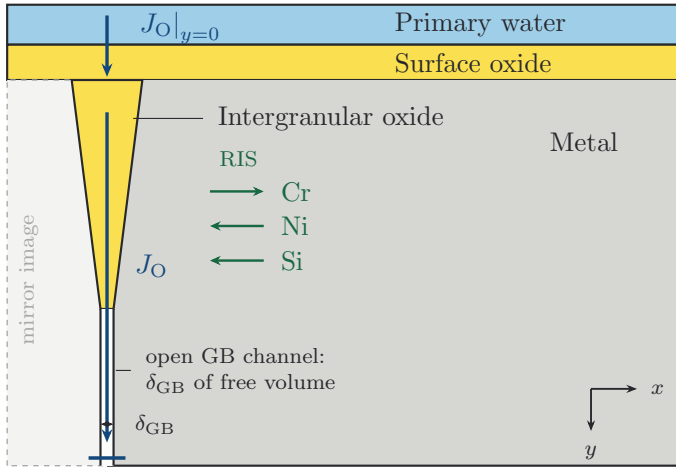
1 Introduction

2 Model Formulation and Calibration

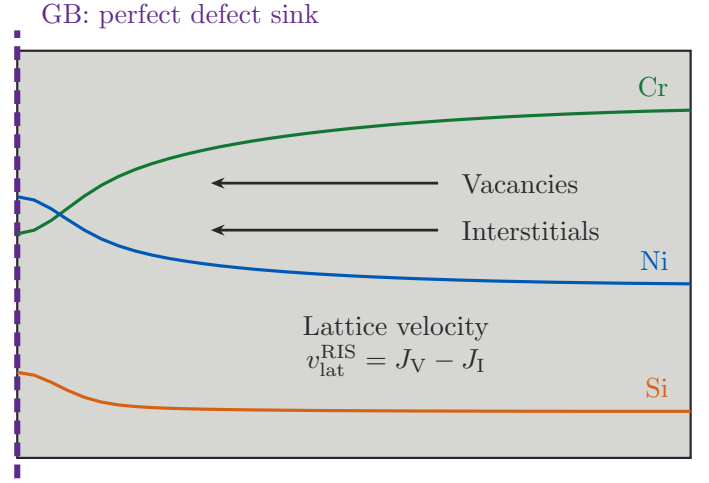
The model consists of two components: a radiation-induced segregation (RIS) model for the evolving GB chemistry and a reduced-dimensional oxidation model for oxygen transport, consumption, and oxide growth. The RIS-predicted Cr, Ni, Si, and Fe concentrations provide the local alloy composition for the oxidation model. Oxygen transport along the GB is solved explicitly, while transport through the oxide and interfacial reactions are treated using a quasi-steady resistance formulation. The oxide composition determines the effective oxygen diffusivity through an effective-medium treatment, which controls the local oxygen consumption and oxide growth rate. The formulations of the two components are described below.

Figure 1 summarises the model. The computational domain, Fig. 1a, is a grain boundary of width δ_{GB} that opens onto the surface exposed to the primary water. Oxygen enters at the mouth through the surface film and migrates along the boundary. While the oxide is thinner than δ_{GB} the boundary retains free volume, which carries a fast oxygen path; once the oxide fills the channel that contribution is lost. Only the half-domain on one side of the boundary is simulated, the other half following by symmetry.

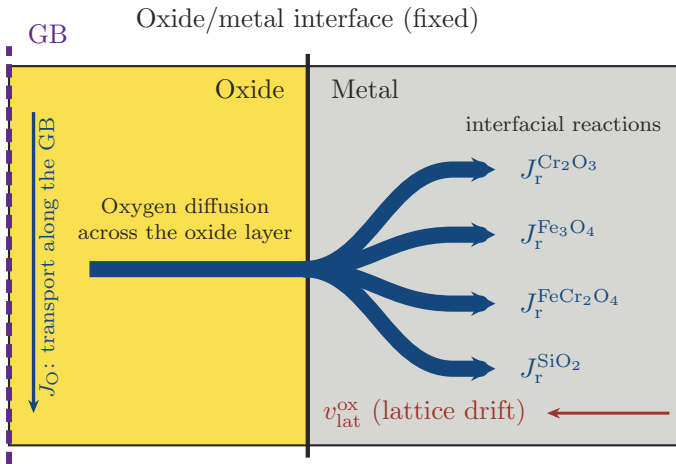
The boundary acts as a perfect sink for the vacancies and self-interstitials produced by irradiation, Fig. 1b. The resulting redistribution depletes Cr



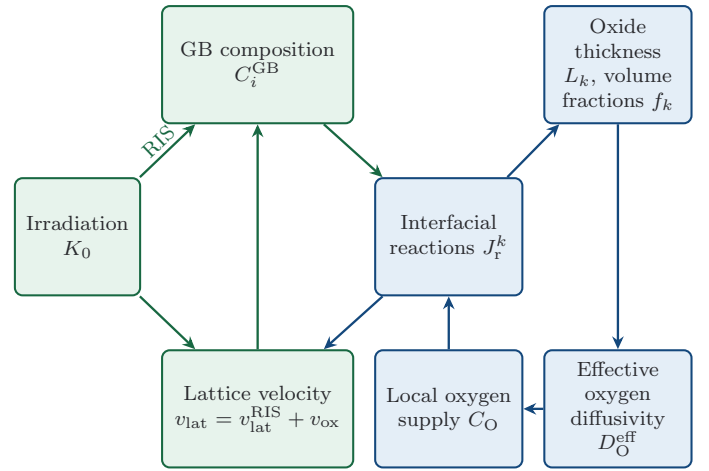
(a)



(b)



(c)



(d)

Figure 1: Overview of the model: (a) physical picture and computational domain, (b) radiation-induced segregation, (c) oxygen transport across the oxide and the interfacial reactions, and (d) coupling between the two components.

and enriches Ni and Si at the boundary, and the imbalance between the two defect fluxes gives the segregation contribution $\mathbf{v}_{\text{lat}}^{\text{RIS}} = \mathbf{J}_V - \mathbf{J}_I$ to the lattice velocity; the profiles shown are calculated, mirrored about the boundary and convolved with a 1 nm FWHM Gaussian. At a given depth, Fig. 1c, the oxygen delivered along the boundary crosses the oxide in quasi-steady state and is partitioned at the fixed oxide/metal interface among the four oxide phases; the metal removed there sets the oxidation contribution v_{ox} . The two components are coupled as shown in Fig. 1d: irradiation fixes the boundary chemistry and part of the lattice velocity, the boundary chemistry controls the interfacial reaction rates, and the oxide that forms feeds back on the oxygen supply through its thickness and its effective diffusivity.

Notation. Alloying-element indices are $i, j \in \{\text{Cr}, \text{Ni}, \text{Si}, \text{Fe}\}$; oxide-phase indices are $k \in \{\text{Cr}_2\text{O}_3, \text{Fe}_3\text{O}_4, \text{FeCr}_2\text{O}_4, \text{SiO}_2\}$. Superscripts V and I denote quantities associated with vacancies and self-interstitials; the superscript GB denotes a value taken on the grain boundary. C is a site fraction, $\mathbf{J}$ a flux in the two-dimensional metal domain, and J a flux on the one-dimensional grain-boundary line. The coordinate x is normal to the boundary and y runs along it.

2.1 Radiation-Induced Segregation Model

The RIS model is based on solving a set of coupled mass-conservation equations for the alloying elements (Cr, Ni, Si, and Fe) and point defects (vacancies and self-interstitials). For each alloying element i , the concentration evolves according to

$$\frac{\partial C_i}{\partial t} = -\nabla \cdot \mathbf{J}_i. \quad (1)$$

The site fractions of vacancies and self-interstitials are governed separately by

$$\frac{\partial C_V}{\partial t} = -\nabla \cdot \mathbf{J}_V + \epsilon K_0 - R C_I C_V, \quad (2)$$

$$\frac{\partial C_I}{\partial t} = -\nabla \cdot \mathbf{J}_I + \epsilon K_0 - R C_I C_V, \quad (3)$$

where C_V and C_I are the vacancy and self-interstitial site fractions, ϵ is the damage efficiency, K_0 is the radiation dose rate, and R is the vacancy–interstitial recombination rate coefficient. The flux of each alloying element

carried by each defect mechanism is

$$\mathbf{J}_i^V = - \sum_j L_{ij}^V \nabla(\mu_j - \mu_V), \quad (4)$$

$$\mathbf{J}_i^I = - \sum_j L_{ij}^I \nabla(\mu_j + \mu_I), \quad (5)$$

with the Onsager coefficients

$$L_{ii}^V = \frac{C_i D_i^V C_V}{k_B T} \left(1 + \frac{1 - f_0^V}{f_0^V} \frac{C_i D_i^V}{\sum_m C_m D_m^V} \right), \quad (6)$$

$$L_{ij}^V = L_{ji}^V = \frac{1 - f_0^V}{f_0^V} \frac{C_i D_i^V C_j D_j^V C_V}{k_B T \sum_m C_m D_m^V}, \quad i \neq j, \quad (7)$$

$$L_{ii}^I = \frac{C_i D_i^I C_I}{k_B T} \left(1 + \frac{1 - f_0^I}{f_0^I} \frac{C_i D_i^I}{\sum_m C_m D_m^I} \right), \quad (8)$$

$$L_{ij}^I = L_{ji}^I = \frac{1 - f_0^I}{f_0^I} \frac{C_i D_i^I C_j D_j^I C_I}{k_B T \sum_m C_m D_m^I}, \quad i \neq j, \quad (9)$$

where D_i^V and D_i^I are the mechanism-specific diffusion coefficients of element i , and f_0^V and f_0^I are the corresponding correlation factors. The net defect fluxes follow from conservation of lattice sites,

$$\mathbf{J}_V = - \sum_i \mathbf{J}_i^V, \quad (10)$$

$$\mathbf{J}_I = \sum_i \mathbf{J}_i^I, \quad (11)$$

and give rise to a lattice velocity

$$\mathbf{v}_{\text{lat}}^{\text{RIS}} = \mathbf{J}_V - \mathbf{J}_I. \quad (12)$$

2.2 Grain-Boundary Oxidation Model

Oxygen transport along the boundary, treated as a slab of width δ_{GB} , obeys

$$\frac{\partial C_O}{\partial t} = - \frac{\partial J_O}{\partial y} - \frac{J_r}{\delta_{\text{GB}}}, \quad (13)$$

$$J_O = -D_O^{\text{eff}} \frac{\partial C_O}{\partial y}, \quad (14)$$

where the effective diffusivity is obtained from an effective-medium mixture of the oxide phases,

$$D_O^{\text{eff}} = \sum_k f_k D_O^k, \quad (15)$$

with f_k the volume fraction and D_O^k the oxygen diffusivity of phase k . Oxygen delivery from the boundary channel to the metal/oxide interface is treated as a quasi-steady series resistance across the oxide of total thickness $\sum_k L_k$,

$$J_r = D_O^{\text{eff}} \frac{C_O - C_O^{\text{int}}}{\sum_k L_k}, \quad (16)$$

where L_k is the thickness of oxide k and C_O^{int} is the oxygen concentration at the interface. The interfacial reaction rates of the four oxide phases are

$$J_r^{\text{Cr}_2\text{O}_3} = k_r^{\text{Cr}_2\text{O}_3} C_O^{\text{int}} C_{\text{Cr}}^{\text{GB}}, \quad (17)$$

$$J_r^{\text{Fe}_3\text{O}_4} = k_r^{\text{Fe}_3\text{O}_4} (C_O^{\text{int}} - C_O^{\text{th}}) C_{\text{Fe}}^{\text{GB}}, \quad (18)$$

$$J_r^{\text{FeCr}_2\text{O}_4} = k_r^{\text{FeCr}_2\text{O}_4} C_O^{\text{int}} \left(\frac{1}{C_{\text{Fe}}^{\text{GB}}} + \frac{2}{C_{\text{Cr}}^{\text{GB}}} \right)^{-1}, \quad (19)$$

$$J_r^{\text{SiO}_2} = k_r^{\text{SiO}_2} C_O^{\text{int}} C_{\text{Si}}^{\text{GB}}, \quad (20)$$

where k_r^k is the interfacial rate constant of phase k and C_O^{th} is the threshold oxygen concentration for magnetite formation. Their sum gives the total interfacial consumption,

$$J_r = J_r^{\text{Cr}_2\text{O}_3} + J_r^{\text{Fe}_3\text{O}_4} + J_r^{\text{FeCr}_2\text{O}_4} + J_r^{\text{SiO}_2}. \quad (21)$$

Equations (16) and (21) constitute the quasi-steady balance between oxygen supply and oxygen consumption,

$$D_O^{\text{eff}} \frac{C_O - C_O^{\text{int}}}{\sum_k L_k} = \sum_k J_r^k(C_O^{\text{int}}), \quad (22)$$

which is solved at every position for the single unknown C_O^{int} ; the supply term decreases and the consumption term increases monotonically with C_O^{int} , so the root is unique. The corresponding rates of metal removal follow from the stoichiometry of each phase,

$$J_r^{\text{Cr}} = \frac{2}{3} J_r^{\text{Cr}_2\text{O}_3} + \frac{2}{4} J_r^{\text{FeCr}_2\text{O}_4}, \quad (23)$$

$$J_r^{\text{Fe}} = \frac{3}{4} J_r^{\text{Fe}_3\text{O}_4} + \frac{1}{4} J_r^{\text{FeCr}_2\text{O}_4}, \quad (24)$$

$$J_r^{\text{Si}} = \frac{1}{2} J_r^{\text{SiO}_2}. \quad (25)$$

2.3 Coupling Strategy and Simulation Protocol

2.3.1 Lattice Velocity and Metal Consumption

The interfacial reactions remove metal atoms only at the grain-boundary plane. Because the metal lattice is incompressible and the boundary is a perfect defect sink, this removal creates no source in the interior: the entire lattice translates towards the interface at the velocity set by the interfacial consumption. Denoting by

$$v_{\text{ox}}(y, t) = J_{\text{r}}^{\text{Cr}} + J_{\text{r}}^{\text{Fe}} + J_{\text{r}}^{\text{Si}} \quad (26)$$

the recession rate of the metal/oxide interface, the total lattice velocity in the metal domain is

$$\mathbf{v}_{\text{lat}}(x, y, t) = \mathbf{v}_{\text{lat}}^{\text{RIS}}(x, y, t) + v_{\text{ox}}(y, t) \hat{\mathbf{e}}_x. \quad (27)$$

The two contributions differ in character. The segregation term is local: it is determined pointwise by the defect fluxes at (x, y) . The oxidation term is not; it is evaluated at the boundary and is independent of x , so a single value applies across the whole domain at a given y . Segregation therefore redistributes the lattice internally, whereas oxidation translates it rigidly, so that the total flux of alloying element i entering Eq. (1) is

$$\mathbf{J}_i = \mathbf{J}_i^{\text{V}} + \mathbf{J}_i^{\text{I}} + C_i \mathbf{v}_{\text{lat}}, \quad (28)$$

subject to the interfacial condition

$$\mathbf{J}_i \cdot \hat{\mathbf{e}}_x|_{x=0} = J_{\text{r}}^i, \quad (29)$$

which removes from the boundary plane the atoms incorporated into the growing oxide.

Equations (28) and (29) collect the three ways in which an alloying element can move. The first two terms are the diffusive fluxes carried by the two defect mechanisms, Eqs. (4) and (5); the third is the drift of the entire lattice relative to the laboratory frame; the interfacial consumption enters not as a bulk term but as an outgoing flux on the boundary plane.

The two contributions to $\mathbf{v}_{\text{lat}}$ have different origins but act identically on every species. $\mathbf{v}_{\text{lat}}^{\text{RIS}}$ arises because the vacancy and interstitial fluxes are unequal, so lattice sites are created and destroyed at different rates; v_{ox}

arises because oxidation consumes metal atoms from the interfacial layer, so the lattice is displaced towards the advancing front. Because the drift term $C_i \mathbf{v}_{\text{lat}}$ is proportional to the local concentration and carries no species selectivity, it does not by itself alter the relative composition; selectivity enters only through the mechanism-specific diffusivities in Eqs. (6)–(9) and through the stoichiometric sinks in Eqs. (23)–(25).

2.3.2 Sequential Irradiation–Oxidation Protocol

The reference experiments irradiate the specimen first and expose it to the primary water environment afterwards. The model reproduces this protocol as two consecutive stages that share the same governing equations and differ only in two switches: the dose rate K_0 and the oxygen boundary condition at the grain-boundary mouth.

In the irradiation stage the specimen is irradiated to the target dose with no oxygen present, $K_0 > 0$ and $C_{\text{O}} \equiv 0$, so that Eqs. (1)–(12) govern the evolution alone and all interfacial reaction rates vanish identically. The stage is integrated to $t_{\text{irr}} = \Phi/K_0$, where Φ is the nominal dose in dpa, and yields the segregated composition profiles $C_i(x, y, t_{\text{irr}})$.

The oxidation stage continues from the final state of the irradiation stage: all concentration fields, including the point-defect populations, are inherited unchanged,

$$(C_i, C_{\text{V}}, C_{\text{I}})|_{t=0+} = (C_i, C_{\text{V}}, C_{\text{I}})|_{t=t_{\text{irr}}}, \quad L_k|_{t=0+} = 0, \quad (30)$$

and the exposure clock is restarted from zero. The dose rate is then set to zero, $K_0 = 0$, so that the inherited defect supersaturation decays by recombination according to Eqs. (2) and (3) without further production.

Simultaneously, the oxygen boundary condition at the mouth of the boundary is switched on. Oxygen ingress is limited by the surface film that grows on the free surface, whose resistance increases as the film thickens; this is imposed as a Robin condition with a time-decaying transfer coefficient,

$$J_{\text{O}}|_{y=0} = h(t) (C_{\text{O}}^{\text{w}} - C_{\text{O}}|_{y=0}), \quad h(t) = \frac{k_{\text{s}}}{\sqrt{t} + \sqrt{t_0}}, \quad (31)$$

where C_{O}^{w} is the oxygen concentration of the bulk water, k_{s} is the surface transfer coefficient, and t_0 regularizes the initial instant. The $t^{-1/2}$ decay reproduces the parabolic thickening of the surface film, so that the cumulative

oxygen admitted through the mouth grows as $t^{1/2}$. The deep end of the boundary is closed, $J_{\text{O}}|_{y=L_{\text{GB}}} = 0$.

Because no point defects are produced once $K_0 = 0$, the inherited supersaturation decays on the recombination timescale, which is short compared with the exposure time; segregation therefore effectively ceases early in the exposure stage, and the composition field that the oxidation model sees is the one established during irradiation. The sequential protocol thus isolates the compositional pathway: any dose dependence of the predicted oxidation depth must be carried by the radiation-induced segregation profile inherited through Eq. (30), since neither radiolysis nor sustained irradiation-assisted transport is present.

2.3.3 Fully Coupled Irradiation–Oxidation Framework

The sequential protocol of Sec. 2.3.2 is a special case of the governing equations rather than a separate model: it is obtained by switching K_0 and the mouth boundary condition on and off in turn. Leaving both switches on throughout, $K_0 > 0$ together with the Robin condition Eq. (31), gives the fully coupled problem in which segregation and oxidation proceed simultaneously, as they do in reactor service. No additional equations or parameters are required; Eqs. (1)–(12) are integrated as a single stage.

In this mode the two processes feed back on each other. Segregation continues to modify the boundary composition while the oxidation front advances, so the composition that the interfacial reactions see evolves in time rather than being fixed at its end-of-irradiation value; conversely, the metal consumed at the front and the associated lattice drift, Eq. (26), alter the concentration field on which the segregation fluxes act. Predictions obtained in this mode are reported in Sec. 3.5; because no experiment in the present dataset imposes concurrent irradiation and exposure, this mode is used only for prediction and takes no part in the calibration of Sec. 2.4.

2.4 Model Parameterization and Calibration

The two components are parameterized and calibrated separately and against separate data. The RIS model is calibrated first; the oxidation model is then calibrated with the RIS model related parameters fixed.

2.4.1 RIS Parameterization

The RIS model requires the mechanism-specific diffusion coefficients D_i^V and D_i^I of the four alloying elements, the correlation factors f_0^V and f_0^I , the recombination coefficient R , the damage efficiency ϵ , and the dose rate K_0 . The dose rate and irradiation temperature are those of the reference experiment. The correlation factors are taken from their crystallographic values for the fcc lattice. The damage efficiency accounts for the fraction of displacements that survive in-cascade recombination and escape as freely migrating defects; its value is that reported for proton irradiation. The vacancy and interstitial migration energies and pre-exponential factors are taken from the literature and adjusted within their reported ranges to reproduce the measured grain-boundary composition profiles across the dose series. Because these profiles are measured before exposure, the RIS calibration is independent of the oxidation model. The resulting values are given in Table 1.

2.4.2 Oxidation Kinetics Parameterization

The oxidation model requires the interfacial rate constants k_r^k of the four oxide phases, their oxygen diffusivities D_O^k , the surface transfer coefficient k_s of the Robin condition Eq. (31), and the magnetite threshold C_O^{th} . The oxide molar volumes and densities used in the thickness update are taken from crystallographic data and held fixed.

The model is calibrated against the dose-dependent grain-boundary oxidation depths of, measured after exposure in simulated PWR primary water at 0, 0.5, and 3 dpa. The predicted oxidation depth is defined as the distance from the boundary mouth to the deepest position at which the total oxide thickness $\sum_k L_k$ exceeds a fixed threshold L_{th} , applied identically at every dose. The free parameters are adjusted to reproduce the measured depths at all three doses simultaneously, using a single parameter set; no parameter is allowed to take different values at different doses. The resulting values are given in Table 2.

Table 1: Parameters of the radiation-induced segregation model.

Symbol	Description	Value	Source
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Table 2: Parameters of the grain-boundary oxidation model.

Symbol	Description	Value	Source
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3 Results

3.1 Radiation-Induced Segregation Profiles

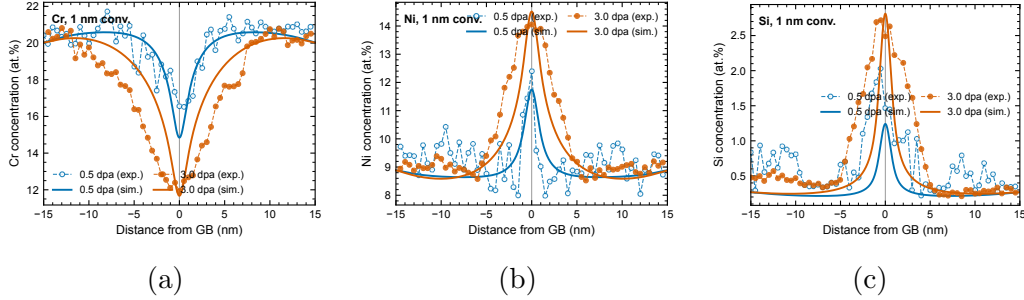


Figure 2: Calculated and measured grain-boundary composition profiles after irradiation to 0.5 and 3.0 dpa: (a) Cr, (b) Ni, (c) Si. Solid lines are calculated profiles convolved with a 1 nm Gaussian; open and filled symbols are the measurements of at 0.5 and 3.0 dpa respectively.

Figure 2 compares the calculated and measured grain-boundary composition profiles after irradiation to 0.5 and 3.0 dpa. The calculated profiles are convolved with a 1 nm Gaussian to account for the finite spatial resolution of the measurements. These profiles are the target of the RIS calibration of Sec. 2.4.1.

Cr is depleted at the boundary and Ni and Si are enriched, at both doses. The boundary Cr concentration falls from a matrix value of 20.4 at.% to 14.9 at.% at 0.5 dpa and to 11.8 at.% at 3.0 dpa, against measured values of 16.2 at.% and 11.8 at.% (Fig. 2a). Ni rises from 8.9 at.% to 11.1 at.% and 14.1 at.% against measured 12.1 at.% and 13.8 at.% (Fig. 2b). Si rises from 0.27 at.% to 1.1 at.% and 2.4 at.% against measured 1.5 at.% and 2.7 at.% (Fig. 2c).

Overall, the RIS profiles show clear dose-dependent changes in both the magnitude and spatial distribution of grain-boundary segregation. The most

pronounced changes are observed in the grain-boundary enrichment or depletion levels, while the characteristic widths of the segregation profiles remain broadly similar across doses. Only modest differences in the full width at half maximum (FWHM) are observed, generally on the order of 1-2 nm, with the most noticeable changes occurring for Cr and Si at 3 dpa. Thus, the dose dependence is primarily reflected in the segregation magnitude rather than substantial changes in the profile width.

3.2 Oxide Composition and Interfacial Ni Enrichment

Figure 3 compares the calculated grain-boundary chemistry with the measurements of **deng**. Because the reported EDX line scans were taken at a single depth on each specimen, the calculated quantities are evaluated at the corresponding depths, measured from the micrographs as 23, 30 and 44 nm for 0, 0.5 and 3.0 dpa.

At those depths the model reproduces the measured dose dependence of the oxide cation composition (Fig. 3a). The calculated Cr content falls from 84 to 65 at.% between 0 and 3.0 dpa, against a measured decrease from 75 to 58 at.%; Fe rises from 15 to 36 at.%, against a measured 24 to 38 at.%. Both trends follow from the Cr depletion established during irradiation: less Cr is available at the front, so the oxide that forms there is correspondingly poorer in Cr and richer in Fe. Ni oxidation is not included in the model, so the calculated Ni content of the oxide is identically zero; the measurements give a few at.% at all three doses, which slightly increase with dose due to the grain boundary positive segregation.

The measured Si content of the oxide is close to zero at every dose, while the calculated SiO₂ volume fraction rises from 1.1 to 25.6% over the same dose range (Fig. 3b). The two are not in conflict. Silica formed at the grain boundary is not retained: the segregated Si is oxidised preferentially and the resulting SiO₂ dissolves into the water, leaving porosity behind in the intergranular oxide (**du**). Its absence from the line scans is therefore expected even where the model predicts that a substantial amount has formed.

The model adopts this behaviour explicitly. All SiO₂ produced is treated as dissolved, and the volume it occupied is retained as free volume available for oxygen transport. The calculated SiO₂ volume fraction is thus a measure of the porosity generated within the intergranular oxide rather than of a phase present in it, and it increases with dose because Si segregation to the boundary increases with dose.

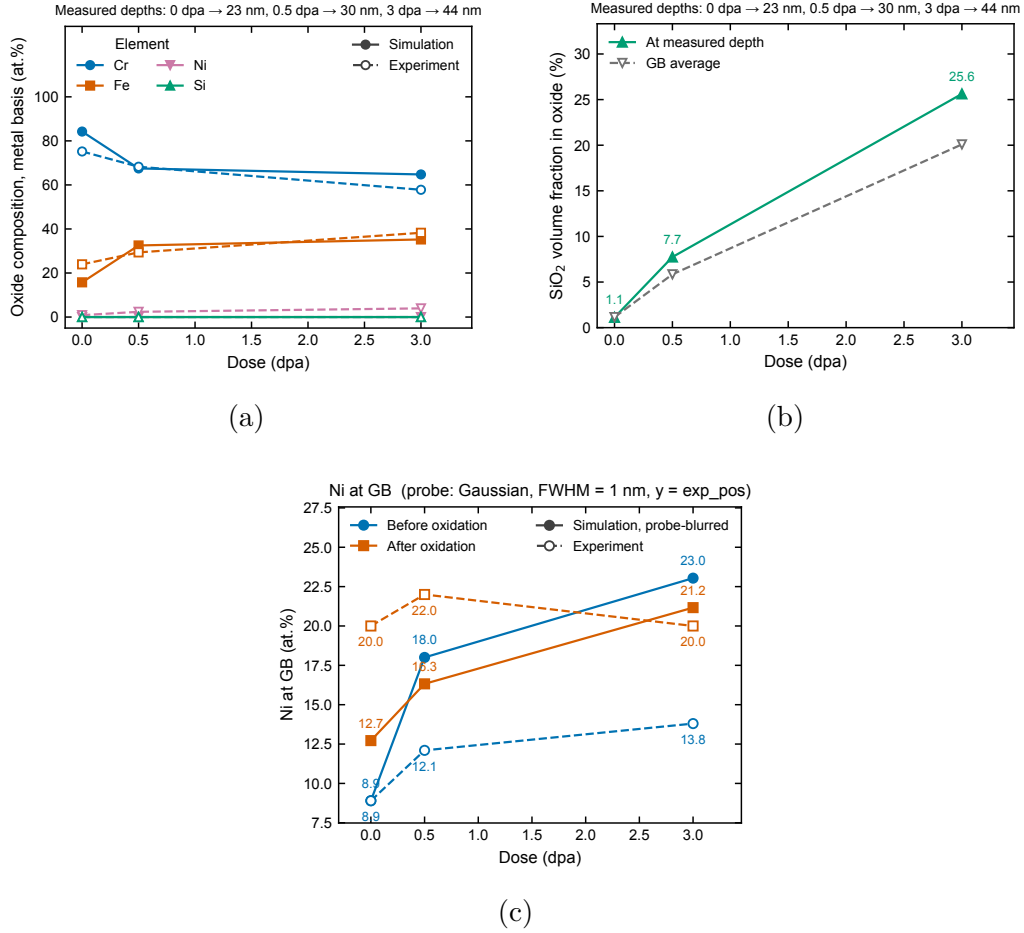


Figure 3: Calculated and measured grain-boundary chemistry as a function of dose. (a) Oxide composition on a metal (cation) basis, evaluated at the depth of the corresponding EDX line scan: 23, 30 and 44 nm for 0, 0.5 and 3.0 dpa. (b) SiO₂ volume fraction in the oxide, both at the same depths and averaged over the oxidized length of the boundary. (c) Ni concentration at the grain boundary before and after exposure; calculated profiles are convolved with a Gaussian of 1 nm FWHM to match the quoted EDX probe diameter and are sampled at the depth of the measurement. Filled symbols with solid lines are calculations, open symbols with dashed lines the measurements of deng.

3.3 Dose-Dependent Grain-Boundary Oxidation Kinetics

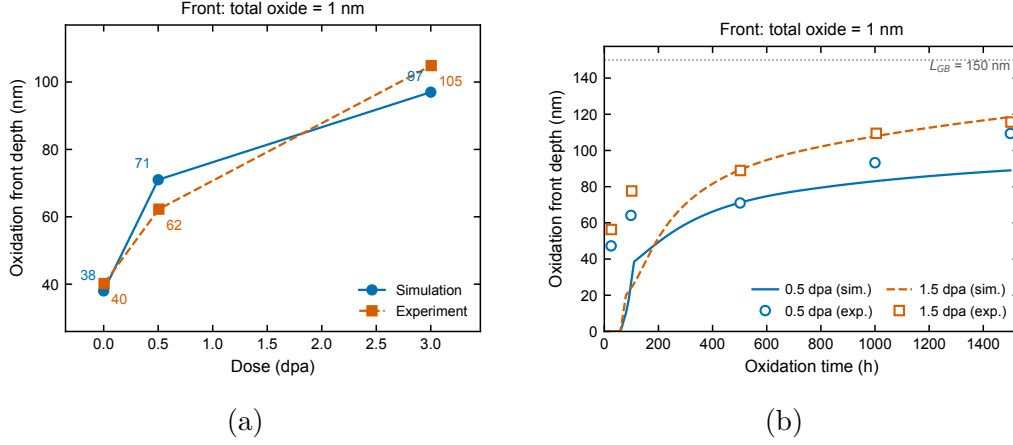


Figure 4: Calculated and measured oxidation front depth, taken at a total oxide thickness of 1 nm. (a) Dependence on dose after the 500 h exposure; these measurements are the ones used to calibrate the oxidation parameters. (b) Dependence on exposure time at 0.5 and 1.5 dpa, obtained with the calibrated parameters held fixed; neither the 1.5 dpa dose nor the exposure times shown enter the calibration. Measurements from **deng**.

- 3.4 Predictions against Independent Experiments
- 3.5 Fully Coupled Predictions under Concurrent Irradiation and Oxidation
- 4 Discussion
 - 4.1 Effect of Radiation-Induced Segregation on Grain-Boundary Oxidation
 - 4.2 Rate-Controlling Transport and Oxide Chemistry
 - 4.3 Implications of Concurrent Irradiation and Oxidation
 - 4.4 Predictive Capability and Model Limitations
- 5 Conclusions